cse 242 hw1

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1

1. (a) the characteristic polynomial for this matrix Σ is

$$\lambda^2 - 41\lambda + 175 \tag{1}$$

which results in eigenvalues of

$$\frac{1}{2}(41 \pm 3\sqrt{109}) \tag{2}$$

setting $(\Sigma - \lambda 1)\, v = 0$, solving for v gives the eigenvectors of λ. this results in $(\frac{1}{10}(3 \pm \sqrt{109}), 1)$
we can check if

$$\begin{pmatrix} 25 & 15 \\ 15 & 16 \end{pmatrix} \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} = \frac{1}{2}(41 + 3\sqrt{109}) \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} \tag{3}$$

and we find that this equation holds.

$$\begin{aligned} \begin{pmatrix} 25 & 15 \\ 15 & 16 \end{pmatrix} \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} &= \begin{pmatrix} 15 + \frac{5}{2}(3 \pm \sqrt{109}) \\ 16 + \frac{3}{2}(3 \pm \sqrt{109}) \end{pmatrix} \\ &= \frac{1}{2}(41 + 3\sqrt{109}) \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} \end{aligned} \tag{4}$$

(b) a plot of the pdf would have 1 std extend 4 units in 1 axis and 5 in the other. the level curve would be an ellipse with some positive correlation between the two axis since the cov is positive.

(c) the direction that preserves the most variance is the one corresponding to the largest eigenvector of the matrix.

the fraction of variance preserved would be $\lambda/41 \approx 0.88$ where 41 is the trace of the cov matrix.

2.

$$
\begin{aligned}
Tr(AB) &= \sum_{i}^{m}(AB)_{ii} \\
&= \sum_{i}^{m}\sum_{j}^{n}(A_{ij}B_{ji}) \\
&= \sum_{j}^{n}\sum_{i}^{m}(A_{ij}B_{ji}) \\
&= \sum_{j}^{n}(BA)_{jj} \\
&= Tr(BA)
\end{aligned} \tag{5}
$$

3.

$$
\begin{aligned}
E[x^TAx] &= E[(\mu + (x-\mu))^TA(\mu + (x-\mu))] \\
&= E[(\mu^TA\mu) + (\mu^TA(x-\mu)) \\
&\quad + ((x-\mu)^TA\mu) + ((x-\mu)^TA((x-\mu)))] \\
&= (\mu^TA\mu) + E[(\mu^TA(x-\mu)) \\
&\quad + ((x-\mu)^TA\mu) + ((x-\mu)^TA((x-\mu)))] \\
&= (\mu^TA\mu) + E[(\mu^TA(x-\mu)) \\
&\quad + ((x-\mu)^TA\mu) + ((x-\mu)^TA((x-\mu)))] \\
&= (\mu^TA\mu) + E[((x-\mu)^TA((x-\mu)))] \\
&= (\mu^TA\mu) + E[A((x-\mu)^T((x-\mu)))] \\
&= (\mu^TA\mu) + tr(A\epsilon)
\end{aligned} \tag{6}
$$

4.

$$E[X] = \int_{-1}^{1} \frac{x}{2}dx = 0 \tag{7}$$

$$E[X^2] = \int_{-1}^{1} \frac{x^2}{2}dx = \frac{1}{3} \tag{8}$$

$$E[X^3] = \int_{-1}^{1} \frac{x^3}{2}dx = 0 \tag{9}$$

$$E[XY] = \int_{-1}^{1} \frac{x^3}{2}dx = 0 \tag{10}$$

Cov is however,

$$E[X^3] - E[X]E[X^2] = 0 \tag{11}$$

since cov is zero, so is correlation but XY are obviously dependent.

2

1. (a)

$$P(d|+) = \frac{P(+|d)P(d)}{P(+)} = P(d|+) = \frac{P(+|d)P(d)}{P(+|d) \cdot P(d) + P(+|\neg d) \cdot P(\neg d)}$$
$$= \frac{1}{100} \tag{12}$$

$$P(+|d) \cdot P(d) = 0.99 \cdot 0.0001 = 0.000099$$

$$P(+) = (0.99 \cdot 0.0001) + (0.01 \cdot 0.9999)$$

$$P(D|+) = \frac{0.000099}{0.010098} \approx 0.0098$$

   (b) this statement doesnt work because it doesnt take in account of the prior probability of having the disease.

   (c)

$$P(d|+) = \frac{P(+|d) \cdot P(d)_{new}}{P(+)_{new}}$$

$$0.99 \cdot 0.0098039 \approx 0.0097059$$

$$P(+) = (0.99 \cdot 0.0098039) + (0.01 \cdot 0.9901961) \approx 0.0196$$

$$P(d|+_{second}) = \frac{0.0097059}{0.0196079} \approx 0.495$$

   (d)

$$\frac{P(+|d) \cdot P(d)}{P(+|d) \cdot P(d) + (1-s) \cdot P(\neg d)} \geq 0.5$$

   $s \geq 0.99990099...$

2. (a) the prosecutor is miscalculating the prior. since the prior is known, the statement that the chance is exactly .99 of being guilty is wrong

   (b) the defender is not acknowledging that having this evidence changes the prior. future evidence will build on this new prior instead of the old one.

   (c) you would need to know the prior probability of the defendant being guilty.

3. (a) maximize

$$P^3((1-P)^2) \tag{13}$$

   we get the max at .6

   (b) instead we maximize

$$P^3((1-P)^2)2P \tag{14}$$

   which gives $P = \frac{2}{3}$

(c) $a = b = 1$ means there were two flips, one heads one tails. when they equal two, there were 4 flips w 2 heads 2 tails. as $a + b$ gets large, it means that there were many flips before and that you are very certain to the probability of heads. $\hat{\theta}_{MAP}$ therefore does not change much given an extra sample point or two.

4.

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$L(\mu, \sigma^2) = \left(2\pi\sigma^2\right)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right)$$

we can take log to make derivatives easier while preserving maximums.

$$l(\mu, \sigma^2) = \ln\left[\left(2\pi\sigma^2\right)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right)\right]$$

solving for μ, we can take σ^2 to be constant,

$$\frac{\partial l}{\partial \mu} = \frac{\partial}{\partial \mu}\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right) = \frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right) = 0$$

gives $\mu = 4$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2}\frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

implies

$$-n\sigma^2 + \sum_{i=1}^{n} (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2$$

$$\hat{\sigma}^2 = \frac{(1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2}{4} = 5$$

5. (a) Likelihood: $P(x_1|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_1-\mu)^2}{2\sigma^2}\right)$

Prior: $P(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(\mu-\mu_0)^2}{2\tau^2}\right)$ where $\mu_0 = 0$ and $\tau^2 = 1$.

$$\log P(\mu|x_1) \propto -\frac{(5-\mu)^2}{2(4)} - \frac{\mu^2}{2(1)}$$

$$\frac{d}{d\mu}\left[-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right] = \frac{5-\mu}{4} - \mu = 0$$

this gives $\hat{\mu} = 1$

(b) we instead get

$$\log P(\mu|X) \propto -\sum_{i=1}^{n} \frac{(x_i-\mu)^2}{2\sigma^2} - \frac{(\mu-\mu_0)^2}{2\tau^2}$$

$$\frac{d}{dx}\log P(\mu|X) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) - \frac{\mu-\mu_0}{\tau^2} = 0$$

we get $\hat{\mu} = \frac{n}{n+4}\bar{x}$ where $\bar{x}$ represents the mean of the samples.

(c) the first term in the $\hat{\mu}$ in part b goes to 1 which means it converges to the mean of x as $n \to \infty$.

6. (a)

$$\begin{aligned}
Var(X) &= \frac{\sum(x-\mu)^2}{N} \\
&= \frac{\sum x^2 - 2x\mu + \mu^2}{N} \\
&= \frac{\sum x^2 - \sum 2x\mu + \mu^2}{N} \\
&= \frac{\sum x^2}{N} - \frac{2\mu\sum x}{N} + \mu^2 \\
&= \frac{\sum x^2}{N} - 2\mu\mu + \mu^2 \\
&= E[X^2] - E[X]^2
\end{aligned} \tag{15}$$

(b)

$$\begin{aligned}
Var(aX+b) &= E[(aX+b)^2] - E[aX+b]^2 \\
&= E[(a^2X^2 + 2aXb + b^2)] - (aE[X]+b)^2 \\
&= E[(a^2X^2 + 2aXb + b^2)] \\
&\quad - [(aE[X])^2 + 2aE[X]b + b^2] \\
&= E[(a^2X^2)] - (aE[X])^2 \\
&= a^2 Var(X)
\end{aligned} \tag{16}$$

(c)

(d)

$$\begin{aligned}
Var(X+Y) &= E[(X+Y)^2] - E[X+Y]^2 && (17)\\
&= E[X^2+2XY+Y^2] - (E[X]+E[Y])^2 && (18)\\
&= E[X^2+2XY+Y^2] - (E[X]+E[Y])^2 && (19)\\
&= E[X^2+2XY+Y^2] && (20)\\
&\quad - (E[X]^2+2E[X]E[Y]+E[Y]^2)\\
&= E[X^2]+E[2XY]+E[Y^2] && (21)\\
&\quad - (E[X]^2+2E[X]E[Y]+E[Y]^2)\\
&= E[X^2]+2E[X]E[Y]+E[Y^2] && (22)\\
&\quad - (E[X]^2+2E[X]E[Y]+E[Y]^2)\\
&= E[X^2]+E[Y^2]-(E[X]^2+E[Y]^2) && (23)\\
&= Var(X)+Var(Y) && (24)
\end{aligned}$$

note the lines 21 and 22 are the lines that require independence. $E[XY] = E[X]E[Y]$

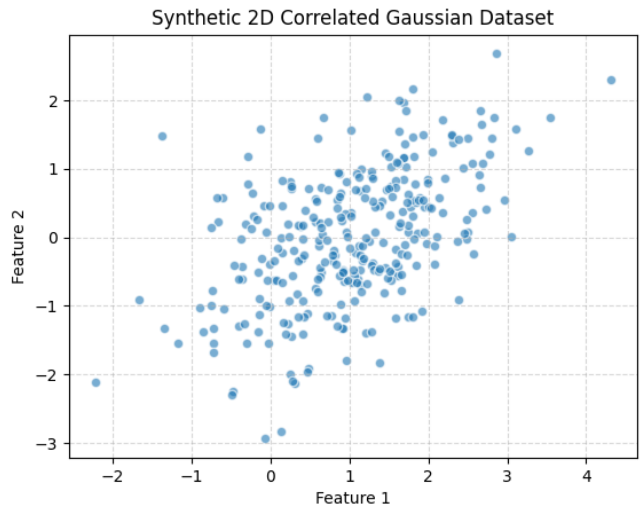
3

1.

$$\begin{aligned}
Var(u^T X) &= E[(u^T X - u^T \mu)^2]\\
&= E[(u^T(X-\mu))(u^T(X-\mu))^T]\\
&= E[u^T(X-\mu)(X-\mu)^T u] && (25)\\
&= u^T E[(X-\mu)(X-\mu)^T]u\\
&= u^T \Sigma u
\end{aligned}$$

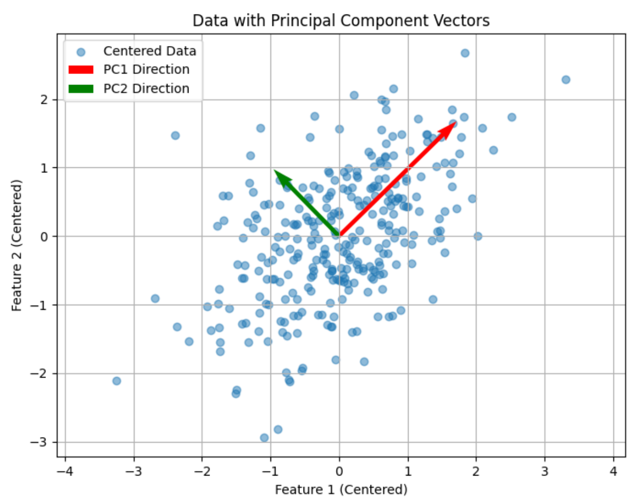
choosing the largest eigenvalues means we get to retain information about the directions that have the largest variance.

2.



the correlation is pretty positive here.

3.




```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('spring_2026\\CSE_242\\synthetic_dataset.csv')

X = df.to_numpy()
```

```
print(f"Dataset Shape: {X.shape}")
print(df.head())

mean = np.mean(X,axis=0)
print(mean)

# center dataset
X_centered = X - mean

#find cov
covariance_matrix = np.cov(X_centered, rowvar=False)

#eig
eigenvalues, eigenvectors = np.linalg.eig(covariance_matrix)

idx = eigenvalues.argsort()[::-1]
eigenvalues = eigenvalues[idx]
eigenvectors = eigenvectors[:, idx]


PC1 = eigenvectors[:, 0]
X_pca = X_centered.dot(PC1)

# --- RESULTS & VISUALIZATION ---

print("Eigenvalues:\n", eigenvalues)
print("\nEigenvectors (Principal Components):\n", eigenvectors)

plt.figure(figsize=(8, 6))
plt.scatter(X_centered[:, 0], X_centered[:, 1], alpha=0.5, label='Centered Data')

# eigenvectors scaled
for i in range(len(eigenvalues)):
    v = eigenvectors[:, i] * np.sqrt(eigenvalues[i]) * 2
    plt.quiver(0, 0, v[0], v[1], angles='xy', scale_units='xy', scale=1,
               color=['r', 'g'][i], label=f'PC{i+1} Direction')

plt.title("Data with Principal Component Vectors")
plt.xlabel("Feature 1 (Centered)")
plt.ylabel("Feature 2 (Centered)")
plt.legend()
plt.axis('equal')
plt.grid(True)
plt.savefig('scatter_plot.png')
plt.show()
```

Eigenvalues: [1.43862294 0.46710326]

Eigenvectors (Principal Components):

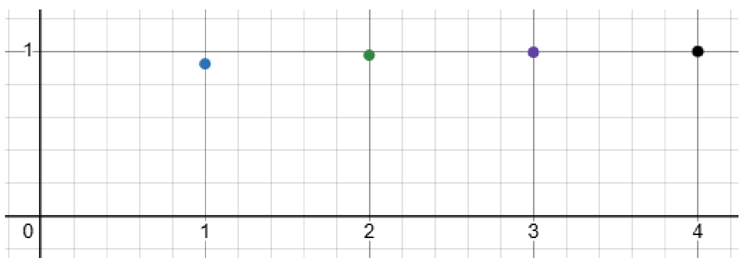
[[0.71516355 -0.69895715]

[0.69895715 0.71516355]]

4. the largest eig captures $\approx \frac{1.5}{1.5+.5} = \frac{3}{4}$ of the total vairance. the second captures $\frac{1}{4}$. the first component captured $\frac{3}{4}$ of the total variance. the higher dim settings are dependent on the application and how accurate you need something to be.

5. I am using sklearn iris.

   eig are $[4.22824171, 0.24267075, 0.0782095, 0.02383509]$

   it takes only 2 components to explain 95% of the varaince

4

1. we can use Q1 in compression after taking data

   we can use Q2 in crime analysis

   we can use Q2 in disease diagnosis

   we can use Q2 in recovering empirical values for collected data in general.

2. i use pypi and documentation for pandas

3. i used gemini fast for pandas and numpy help.

4. Im not sure how long it took since i didnt look ahead but the question part took 2.5 hours. if i had to guess the first parts took longer.